



# Minimizing effects of ‘liability of foreignness’: Response strategies of foreign firms in the United States

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## ABSTRACT

While anecdotal and research evidence exists supporting the difficulties faced by foreign firms in host nation environments due to liability of foreignness, it is clear that many foreign firms have been successfully operating in the U.S. over the years. This study seeks to understand the strategies foreign firms use to cope with liabilities of foreignness in an alien environment and compete successfully with domestic firms, specifically through boundary spanning. Using a sample of 3861 firms in the U.S., we find that foreign firms on the average underperform compared to domestic firms. We also find these firms take a differing strategic posture to cope with the disadvantages of being a foreign firm compared to domestic rivals. Multiple mediation models indicate that once this strategic posture of foreign firms is controlled for, performance differentials do not exist between foreign and domestic firms.

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## 1. Introduction

“...foreign companies blundered by buying control of weak syndicates and hiring second-rate underwriters to run them. It helps to know which underwriters drink too much at lunch... foreigners would be very hard pressed to know the intricacies...” (Hagerty & Oster, 2002: p. A5).

Scholars working in the field of international business, starting with Hymer (1976) and Kindleberger (1969) have articulated that foreign firms face disadvantages while doing business abroad. This concept has been referred to as liability of foreignness (LOF) and has been defined as “all additional costs a firm operating in a market overseas incurs that a local firm would not incur” (Zaheer, 1995, p. 343). The original work of Hymer has been expanded by later researchers who have attributed liability of foreignness to a set of interrelated factors. Eden and Miller (2001)

use the metaphor of “stranger in a strange land” (p. c4) to differentiate the three types of additional costs faced by a foreign firm. They point out that the “stranger” is someone locals do not know, leading to discrimination hazards. According to Hymer (1976, p. 35), the “stigma of being foreign” will persist even after the firm has set up operations in the host country, wherein domestic firms could receive preferential treatment from customers, suppliers, and the local government. Being in a “strange land” leads to unfamiliarity hazards: foreign firms would lack quality information about local markets due to the lack of understanding of local environment and culture. Second, foreign firms face increased administrative expense due to costs of travel and transportation over various time zones, as well as increased setup, coordination, and monitoring costs due to operations being spread across foreign lands (Hitt, Hoskisson, & Ireland, 1994). Finally, changes in currency, economic, and political factors in host countries create higher risk for foreign firms operating in an alien environment, as compared to local firms (Bae & Jain, 2002; Hymer, 1976; Zaheer, 1995).

Several empirical studies exist that provide support for the underlying notion of liabilities of foreignness. De

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Young and Nolle (1996) found that foreign-owned banks operating in the U.S. were less profit-efficient than local U.S. banks. In another study of global banks in 13 countries by Miller and Parkhe (2002), average levels of efficiency scores for foreign banks were lower than their local counterparts. A similar pattern of results was reported by Miller and Richards (2002) in the European Union. While these studies focused on efficiency of banks, Zaheer (1995) and Zaheer and Mosakowski (1997) tested the notion of LOF on foreign currency trading rooms, finding strong support in terms of lower profitability and higher failure rate by foreign-owned trading rooms. In a study done by Mezias (2002a, 2002b) on administrative capabilities of foreign firms in an alien environment, it was found that foreign subsidiaries operating in the United States faced significantly higher labor lawsuit judgments in federal and state courts compared to a matched sample of U.S.-owned firms.

Given the existence of LOF, scholars working on this topic have offered many suggestions to mitigate its effects on foreign firms. Eden and Miller (2001) argue that the entry mode choices should be made to reduce the effects of LOF. Using case studies of Taiwanese firm entry strategies in Europe, Chen (2006) validates this notion. More empirical support for this type of rationale in the context of LOF can be found in Luo, Shenkar, and Nyaw (2002), in which they report that foreign firms locating in China use a combination of offensive (e.g., networking, legitimacy improvement, etc.) and defensive strategies (contracts, *guanxi*, parents control, etc.) to mitigate the effect of LOF. In a study of the Canadian auto industry, other types of offensive strategies are presented by Eden and Molot (2002), wherein first movers use firm specific advantages to become “insiders” and create entry barriers for later arrivals. The importance of exploiting firm specific advantages is highlighted by Nachum (2003), wherein she shows how these resources can be used to outperform local firms in the financial services industry. Haiyang, Griffith, and Ru (2006) integrate entry mode choice with firm specific advantages and foreign firm entry strategies into the Chinese market, and also posit they are influenced by the degree of LOF.

While extant research serves as an effective foundation for managing LOF through entry mode choice and deployment of firm specific advantages, this study seeks to contribute to the literature by focusing on understanding how foreign firms relate to the alien environment through boundary spanning (Scott, 1998). Boundary spanning refers to “...those roles that involve procuring resources and disposing of outputs, relating the organization to its larger community, and adapting the organization to the future by gathering information about trends ...” (Scott, Mitchell, & Birnbaum, 1981: p. 244). This topic deserves attention, as foreign firm survival and success will be enhanced by managing and learning from the environment, even in instances of appropriate entry mode choice and deployment of firm specific advantages. Therefore, this study seeks to add to the LOF literature by studying the strategic posture of foreign firms. It focuses on a single industry in the U.S. to understand the response mechanisms of foreign firms. By limiting the sample to one type of

industry rather than a cross-section of industries, it serves as a control for industry effects and, more importantly, it will allow for usage of industry specific measures in the comparison between domestic and foreign firms. Mezias (2002a, 2002b) specifically calls for this type of research design to identify liabilities of foreignness (p. 273, Table 1).

Among the varied industry choices, we chose the insurance sector for three reasons. First, the insurance industry belongs to the service sector and is highly regulated compared to most industries in the U.S. Both of these factors are likely to potentially increase LOF for foreign firms, thereby providing an ideal sample for such a study. Second, performance data for these firms are available at the business unit level. This allows us to study these firms without the restraints imposed by traditional financial reports, which largely present consolidated corporate data. Third, we wanted an industry where there is a strong market presence by foreign firms in the U.S., to ensure an adequate sample for this study. We found that U.S. affiliates of foreign insurance-related companies conducted business to the tune of \$111.7 Billion, controlled assets of \$1,127 Billion and had 119,000 U.S. employees during the year 2004 (Anderson & Zeile, 2006). Therefore, to test the study hypotheses, we use regression models comparing U.S. insurance operations of foreign and domestic firms in the insurance industry. Additionally, using multiple-mediation regression models, we seek to understand the impact of these choices on financial performance of firms. Specifically, this study will serve to increase our understanding of strategies of foreign firms in the U.S., and thus generally our understanding of the coping mechanisms of firms in an alien environment.

## 2. Conceptual background and hypothesis development

Strategy literature calls for firms to adapt to the environment to compete successfully. A similar logic has been articulated in LOF literature (Luo & Tan, 1998; Petersen & Pedersen, 2002; Sethi & Guisinger, 2002), where it is argued that a foreign firm operating in an alien environment with inherent disadvantages relative to local firms will take a proactive role to minimize its disadvantages and potentially gain competitive advantage. In this study, we borrow from boundary spanning literature to argue that foreign firms will seek to enhance their ability to learn from alien environments through effective information processing and by transmitting positive information to this environment. Learning in the foreign environment is critical. Petersen and Pedersen (2002) in their study of 494 foreign firms found that willingness to learn and adapt in the local environment enhances the ability of a firm to overcome LOF. Conceptually, using the boundary spanning logic allows this study to incorporate a holistic perspective (i.e., rational, natural or open systems perspectives) and bring in the multiple dimensions of the organizational interface with environment (Galbraith & Kazanjian, 1986; Thompson, 1967). This perspective is also inclusive of the well articulated and accepted institutional perspective in the LOF literature (Kostova & Zaheer, 1999; Zaheer & Mosakowski, 1997) and hence complementary to extant literature.

Scholars have studied boundary spanning roles at the individual (e.g., Perrone, Zaheer, & McEvily, 2003) and organizational levels (e.g., Scott, 1998). However, this study's focus is at the organizational level and specific to the role of managing the interface between organization and environment. In such instances, two roles are primary: information processing and external representation. The first role is in helping the organization learn and perform according to variation in the environment, while the second role helps maintain legitimacy of the organization and its activities by transmitting favorable information (Aldrich, 1979). Empirical support for the importance of such roles in the industry context of this study (i.e., financial services) is provided by Jemison (1984). His findings assert two roles to be critical: internalizing information into the organization about changes in the environment, and transmitting information into the environment that presents the organization favorably.

While boundary spanning roles are important for domestic and foreign firms, the need for managing this task effectively is magnified in the case of foreign firms, as these firms face the disadvantages of LOF. They include the costs associated with distance costs due to unfamiliarity or lack of roots in the local environment costs due to lack of familiarity and economic nationalism and potential restrictions from their home country environment (Zaheer, 1995: p. 343). Therefore, when a foreign firm operates in an alien market, its chances of survival and success in the alien environment will be influenced by its ability to handle these two roles. These two roles allow foreign firms "...to achieve a compromise between organizational policy and environmental constraints ..." and "...to choose strategic moves to overcome constraints..." (Aldrich & Herker, 1977: p. 221). Therefore, in the following paragraphs we develop four hypotheses on the strategic posture of foreign firms relative to domestic firms that help these firms internalize and transmit information in an alien environment despite the disadvantages of LOF.

## 2.1. Hypothesis development

One of the factors that create LOF is costs due to lack of familiarity in the local environment (Zaheer, 1995). This lack of familiarity requires foreign firms to collect information which is already known to local firms (Hennart, Roehl, & Zeng, 2002). On the other side of the coin, there is a lack of awareness by customers and regulators leading to discrimination hazards (Zaheer, 1995) and time compression diseconomies (Markides & Williamson, 1994) during market entry. One way for foreign firms to counter this disadvantage is by operating in a number of product segments in local markets. Two motivations exist for doing this. First, by having a greater number of product choices, the foreign firm increases its odds of offering the customer a product which is desirable compared to a local firm in spite of discrimination hazards and other potential judgmental errors caused by LOF. Second, increased product variety will allow the foreign firm to gain exposure in underserved segments of the local markets, allowing for growth with relatively less competition. Previous studies also support the notion that such

within-business diversification influences firms' survival and ability to cope with environmental dynamism (Stern & Henderson, 2004). Additionally, they bring in scope-related advantages such as asset mass efficiencies (Markides & Williamson, 1994).

**Hypothesis 1.** Other things being equal, foreign-owned firms will have greater product variety in the U.S. compared to domestically owned firms.

Foreign firms do not have strong local ties which could lead to strong sales in one region, as they lack roots in the local environment (Zaheer, 1995). One option is that they could spread themselves across the host country, thereby exposing themselves to a greater number of customers. Since legitimacy is cognitive in nature (e.g., George, Chattopadhyay, Sitkin, & Barden, 2006; Hannan & Freeman, 1984), familiarity with an organization leads to greater credibility and acceptance of its existence. Operating in many geographic regions also allows the creation of a "local network" and thus the tapping of local information and connections which a foreign firm does not have access to (Luo et al., 2002). One can argue, however, that operating in multiple geographic regions may increase LOF for foreign firms. We acknowledge this is a possibility, as foreign firms can reduce LOF by focusing on one region. In such instances, foreign firms will lack scale and be handicapped in operations, having no gains in specialization due to inefficiencies in offering support to customers, claims processing, and in other infrastructural issues relating to services (Mayers & Smith, 1990). Second, by focusing on a narrow region, foreign firms increase volatility in losses and thereby increase bankruptcy risk (due to a lack of risk pooling). For instance, an insurer operating in narrow regions could have financial statements devastated by a single event (e.g., Hurricane Katrina), thus threatening its very survival. Third, insurance firms, as part of their regular operations, resell their policies to other reinsurance firms or use third party capital market risk transfer options such as alternative risk transfer (ART) or catastrophic risk transfer (CRT) mechanisms. When insurance firms lack scale, they may not be as attractive to supplier firms of such specialized services. Thus, firms which provide such services would not offer the best rates and would receive unfavorable rates or second tier treatment. While LOF could be decreased by focusing on narrow markets, in this instance foreign firms would face tradeoffs in legitimacy, information and significant cost and operational penalties. Therefore, in line with the above assertions and boundary spanning rationale advanced in the theoretical section, we propose:

**Hypothesis 2.** Other things being equal, foreign-owned firms will sell products in more states in the U.S. compared to domestically owned firms.

Foreign firms face disadvantages due to a lack of understanding of local culture and underlying market patterns while operating in an alien land (Zaheer & Mosakowski, 1997). Local firms have advantages in this regard. They are likely to have a better understanding of specific market needs due to cultural ties and historical

evolution in the local environment. Additionally, local firms may be socially embedded and well-connected with unaffiliated local organizations giving access to market niches. One of the responses of foreign firms to this situation would be to operate as an “intracorporate network” (referred to henceforth as “group affiliation”), wherein several firms operate within a unified corporate identity<sup>1</sup> (Inkpen & Tsang, 2005). Such groups are a common phenomenon in the insurance sector, where about two-thirds of firms are affiliated to insurance groups, although individual members are separate corporations (Cummins & Sommer, 1996). Affiliation in a group provides opportunities for developing new connections in alien environments and for sharing resources with other members, thereby reducing replication and increasing efficiency in operations (Elango & Pattnaik, 2007). Additionally, in the event of liquidity problems (in times of calamity or disaster) a member has a ready option to borrow from other group members (Colquitt, Sommer, & Godwin, 1999). Examples of shared activities include common branding, infrastructural services, investment departments and financial strategies in lending and borrowing (Cummins, Phillips, & Smith, 2001). By operating as a network, all firms belonging to the group optimize acquisition of such knowledge, as local business and tacit knowledge can be acquired from other members in the network. For instance, member firms could tap into and learn from current activities taking place at business units within their networks to refine and optimize operations. While such learning is likely to offer greater benefit in overcoming LOF when other group members have lengthy experience in the host nation, such learning would be useful even in instances when group firms have limited experience due to shared learning. By being part of a group, information about changes in the host country environment can be internalized into the organization more effectively and efficiently (Jarillo, 1988). The occurrence of foreign firms learning from other group members during international expansion has been reported in the literature (Chang, 1995; Guillen, 2003). Moreover, by being part of a group, they gain from “legitimacy spillovers” (Kostova & Zaheer, 1999), giving them greater credibility in dealing with external stakeholders.

**Hypothesis 3.** *Other things being equal, foreign-owned firms are more likely to be affiliated with a business group (insurance) in the U.S. compared to domestically owned firms.*

Foreign firms operating in an alien environment face a lack of legitimacy since they do not possess roots in the host nation. This happens because the host country environment has less information with which to judge foreign firms (Kostova & Zaheer, 1999). In these circumstances, these firms can reduce discrimination hazards by transmitting into the environment information that presents the organization favorably (Jemison, 1984; Kostova & Zaheer, 1999). One approach for an organization to seek legitimacy is to use institutional legitimating

devices that enhance public trust (Van de Ven & Garud, 1989) and by becoming isomorphic with the institutional environment (Eden & Molot, 2002). For instance, in the case of the insurance industry, foreign firms will use a more conservative (i.e., higher) level of capitalization relative to potential outstanding claims or obligations in their U.S. operations. This is possible because firms expanding internationally usually have greater organization resources and capabilities and would be able to secure necessary capital before such expansion. Such higher capitalization offers two advantages. First, a higher level of capitalization will serve as slack in the event of managerial errors committed due to LOF in an alien land. Second, it will allow foreign firms to secure more favorable ratings through locally reputable independent third-party rating firms which rate the insolvency risk through adequacy of capital. Such favorable ratings would serve as a legitimizing tool and as a valuable benchmark for external stakeholders (i.e., customers and regulators) and reduce the perceived risk of dealing with a foreign firm.

**Hypothesis 4.** *Other things being equal, foreign-owned firms will have higher third-party ratings in the U.S. compared to domestically owned firms.*

### 3. Research methodology

#### 3.1. Data sources

This study used two sources of data, namely the National Association of Insurance Commissioners (NAIC) database and Best's Rating guide. NAIC was created in 1871 as the organization of insurance commissioners in the United States [for additional information, see <http://www.naic.org/>]. Insurance firms report data to NAIC as a part of their overall fiduciary requirements, because regulatory approval (by state) is required for insurance operations in the U.S. The data collected by NAIC is deemed to be highly reliable and has been used in academic insurance research (e.g., Cole & McCullough, 2006; Choi & Weiss, 2005; Cummins & Phillips, 2005) and by regulators (including state legislators) for financial and market conduct regulation. A key advantage of using the NAIC data for a study is that it provides operational and financial data at a business unit level, thereby minimizing confounds (e.g., internal transfer pricing effects, firm specific variances in the grouping of business units, etc.) that can be introduced in aggregate reporting of performance in 10 K reports. The second data source for this study, Best's Ratings, have been published by A. M. Best Company since 1899 and are widely used in research (e.g., Browne, Carson, & Hoyt, 1999; Pottier & Sommer, 1999; Viswanathan, 2006) and accepted as a reputable source for information on insurance companies. Many insurance companies voluntarily report such ratings in their promotional information as a validation of their financial soundness.

#### 3.2. Sample selection

We initially started with data from the NAIC database. Our first step was to identify domestic and foreign insurers.

<sup>1</sup> However, each of these units is required to maintain an independent balance sheet and income statements for regulatory purposes.



Foreign insurer is defined as one with a foreign (non-United States) person or entity directly controlling 100% and reporting to NAIC. We quickly noted many U.S. (and non-U.S.) firms had subsidiaries domiciled in tax-haven countries, thus showing up as foreign companies. This is a common practice by insurance companies due to greater regulatory freedom and taxation policies in those countries. Since this study's key research question is on liabilities of foreignness (which may not apply to American firms located in tax-havens) we decided to eliminate all companies based in countries which are commonly known as tax-havens (i.e., Bermuda, Cayman Islands, Panama, and Trinidad and Tobago). Later, we identified all the companies which have data for each of the study variables in the NAIC database and Best's Rating guide for the years 2000 through 2004<sup>2</sup>. This resulted in a sample of 3789 domestic firms and 51 foreign firms<sup>3</sup>, giving us a total sample of 3861 firms. The following subsection describes the operationalization of the variables pertaining to each of the hypotheses. A later subsection explains the control variables used in the testing of the study models.

### 3.3. Variable operationalization

Product variety is operationalized using a variation of the Herfindahl Index. Product variety is calculated as  $1 - \sum p_i^2$ , where  $p_i$  represents the percentage of an insurer's net premiums written on product line  $i$  in the U.S. Similarly, we measure geographic dispersion using the Herfindahl Index as  $1 - \sum s_i^2$  where  $s_i$  represents the percentage of an insurer's direct premium written on state  $i$ . In both measures, we have a score of 0 (zero) when an insurer is active only in one product (or one state). This score will increase in value with increasing number of products (or states). Similar application of the Herfindahl Index measures has been practiced in previous research (Mayers & Smith, 1990). Insurance firms can either exist as a single independent firm with no group affiliation or as a firm affiliated to an insurance group (i.e., intracorporate network). A majority of insurance firms operate as members of business groups with common ownership ties, whereas only about one-third of firms operate as independent firms. Though group members have a common ownership, individual companies are separate corporations. Within each group a majority of group members are other insurance firms along with a few financial services firms (e.g., leasing, investing, banking). This feature is unique to the insurance industry. Such arrangements allow for sharing of resources, brand-names and financial strategies, while at the same time providing for a separation/segregation of risk profiles of individual units. Additionally, these affiliations allow group members an important option of securing additional financial capital easily from each other in times of need (i.e., a liquidity crisis), a contingency any insurer needs to plan for. Consistent with past research in the insurance literature on this topic (e.g., Colquitt et al., 1999; Cummins & Sommer, 1996; Cummins et al., 2001; Krishnaswami & Pottier, 2001), we measure the

group affiliation through a dummy variable where firms associated with an intracorporate insurance group were classified 1, and 0 otherwise.

To proxy third party ratings, we use the previous year's [i.e.,  $(t - 1)$ ] Best's capital adequacy ratio (CAR) from Best's Key Ratings Guide (1995–2005). This ratio deals with the ability of an insurer to meet promises to its customers (i.e., can the insurer hold good on its obligations to its policyholders when the need arises). This ratio is measured as insurer's *adjusted surplus* divided by the *required capital*, where: *adjusted surplus* is reported surplus<sup>4</sup> minus adjustments to correct for off-balance sheets items and unearned premiums, losses or assets; and *required capital* is the required level of capital to support seven risk categories (i.e., fixed income, equity, interest rate, credit, losses, premiums and business) faced by insurance firms. The efficacy of this particular measure is that it corrects for many items which are usually not factored in the computations of the ratios directly from income statements and balance sheets, thereby allowing for a reasonable comparison across firms. CAR scores can vary from 0 to 1000, and as benchmarks, an insurer with a score of 200 or more is deemed to have relatively strong balance sheet, profitability, etc. to cover for future losses, whereas a score of 100 would indicate a firm having "adequate" capitalization to cover losses. This ratio should be of critical concern for customers (i.e., insurance policyholders), since unlike other industries, advance payments are made by customers to an insurer based on contingent payment in a particular situation at a later date subject to policy triggers and caveats. These ratios are generically referred to as capitalization tests and used for inferring the strength of the company (Myhr & Markham, 2004). Since profitability measures influence CAR scores, in the regression models, previous year's  $CAR_{(t-1)}$  values are used [i.e., regressed with current year's performance<sub>(t)</sub>] to avoid any potential violation of regression assumptions.

### 3.4. Control variables

The regression model incorporates eleven control variables (excluding year dummies) that were selected based on past research, and the reason for inclusion of each can be summed up as follows. *Cultural distance* is included to control for the fact that firms from similar cultures to that of the U.S. would face fewer problems adapting to the U.S. environment (and thereby lesser LOF) compared to firms from more dissimilar cultures. *Age* of the firm needed to be controlled for, as it has been argued that older firms will be able to reduce the impact of LOF due to experience (Calhoun, 2002; Mezas, 2002a, 2002b). *Firm size* is included, as size has been found to share a relationship with performance (Lai & Limpaphayom, 2003). Insurers can differ in their focus by choosing to operate either in personal or commercial lines. This choice of lines will lead to a different emphasis (and costs) on marketing, underwriting, etc., and therefore we control for *Business Line* in

<sup>2</sup> With the exception of CAR values, as they lagged by a year. Hence the data for CAR values are from 1999–2003.

<sup>3</sup> From 11 countries: Australia, Belgium, Canada, Denmark, France, Germany, Japan, Korea, Spain, Switzerland, and the United Kingdom.

<sup>4</sup> This term is used in the insurance industry to accommodate stock firms and mutual firms (since mutual firms do not issue stock). Surplus (referred to as policyholder's surplus) is net worth of a firm under statutory accounting (Myhr & Markham, 2004).

the models (Fiegenbaum & Thomas, 1990). *Investment* is included, as one of the business activities of an insurer is investment, which affects its performance. Following Browne, Carson, and Hoyt (2001), we include the percentage of the investment portfolio invested in stocks as a measure of an insurer's investment activities. *Price* and *Expense ratio* are controlled for to incorporate the particular characteristics of product segment in which the firm operates. For instance, product lines with higher loss rates will have higher insurance rates, and firms in these segments are likely to incur higher costs servicing such customers (Myhr & Markham, 2004). *Price* is measured as the inverse of loss ratio, consistent with previous research practice (Elango, 2003). *Leverage* is included, as it provides a source of capital but also subjects the firm to increasing levels of financial risk, which in turn reduces the performance of the firm (Colquitt & Hoyt, 1997; Sommer, 1996). *International ratio* is included to control for the extent of operations outside U.S. markets, as this study's focus is on liabilities in the host country market (i.e., U.S.). *Organization type* is included, as insurers are structured as stock and mutual firms. Previous research has argued that the pressure to maximize performance is far less in a mutual firm, since the role of owner and policyholder functions are merged (Colquitt et al., 1999). The insurance industry traditionally employs two types of product distribution, the independent and exclusive agency systems. The costs and efficiency of each system vary (Berger, Cummins, & Weiss, 1997; Regan, 1997) and therefore the type of *Distribution system* needs to be controlled for. Table 1 provides the operationalization

procedure of these variables and data sources for this study. The variables were operationalized using commonly prescribed practices in insurance literature and hence are not specifically elaborated (Myhr & Markham, 2004). Descriptive statistics of the variables used in the study, along with correlation values, are presented in Table 2. A review of the correlation values between the various independent variables indicated that the likelihood of multicollinearity invalidating the study findings was minor. Additionally, in each of the regression models, variance inflation factor (VIF) scores were checked to ensure that assumptions of the models tested are not violated. The following section presents the statistical results of this study.

#### 4. Results

The theoretical argument made in this paper is built on the premise that foreign firms face disadvantages in alien environments relative to home country firms. While many studies support this notion, one study reports contrary evidence (Nachum, 2003). Therefore, though no explicit hypothesis was stated, we felt it would be critical to first confirm if foreign firms underperform domestic firms. We measured performance using *return on assets* (ROA), as it is a commonly used measure by managers, regulators, rating agencies and investors. ANOVA comparisons between the foreign and domestic firms (means of .382 and .441, respectively) indicate that on the average foreign firms underperform domestic firms ( $F = 5.432$ ,  $p < .02$ ). To ascertain these performance differences further, we ran

**Table 1**  
Variable operationalization and data source

| Variable              | Operationalization                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Product variety       | $1 - \sum p_i^2$ where $p_i$ represents the percentage of an insurer's net premium written on product line $i$ . <i>Data Source:</i> NAIC database                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Geographic dispersion | $1 - \sum s_i^2$ where $s_i$ represents the percentage of an insurer's direct premium written on state $i$ . <i>Data Source:</i> NAIC database                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Group affiliation     | Dummy variable = 1 if it belongs to a intracorporate group, 0 otherwise. <i>Data Source:</i> Best's Key Rating Guide                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Third party ratings   | Third party ratings are proxied through the previous year's [i.e., $t - 1$ ] Best's capital adequacy ratio (CAR). This ratio is an insurer's <i>adjusted surplus</i> divided by the <i>required capital</i> , where <i>adjusted surplus</i> is reported surplus minus adjustments to correct for off balance sheets and unearned premiums, losses or assets; and <i>required capital</i> is required level of capital to support seven (fixed income, equity, interest rate, credit, losses, premiums & business) risk categories faced by insurance firms. <i>Data Source:</i> Best's Key Rating Guide |
| Foreign dummy         | Dummy variable = 1 for an insurer who is 100% foreign owned, 0 otherwise. <i>Data Source:</i> NAIC database                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Cultural distance     | Using the operationalization of Kogut and Singh (1998), cultural distance is defined as the sum of the absolute values of the differences between the host country and the U.S. with respect to the four dimensions of culture (i.e., power distance, uncertainty avoidance, individualism, and masculinity/femininity) identified by Hofstede (1980)                                                                                                                                                                                                                                                   |
| Age                   | Data reporting year—year of incorporation. <i>Data Source:</i> Best's key rating guide. <i>Data Source:</i> NAIC database                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Size                  | $\ln(\text{assets in millions})$ . <i>Data Source:</i> NAIC database                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Business line         | Dummy variable = 1 if it operates in commercial lines, 0 otherwise. <i>Data Source:</i> NAIC database                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Investment            | [Assets in common stocks]/[total assets]. <i>Data Source:</i> NAIC database                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Price                 | Inverse of loss ratio; where loss ratio = [Incurred losses + Loss adjustment expenses]/earned premiums. <i>Data Source:</i> NAIC database                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Expense ratio         | Expenses incurred/earned premiums. <i>Data Source:</i> NAIC database                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Leverage              | Liabilities/[a firm's surplus, common stock, and preferred capital stock]. <i>Data Source:</i> NAIC database                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| International ratio   | Direct premiums written in international markets/total premiums (expressed as a percentage). <i>Data Source:</i> NAIC database                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Organization          | Dummy variable = 1 if stock company, 0 otherwise. <i>Data Source:</i> Best's Key Rating Guide                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Distribution system   | Dummy variable = 1 if independent agency firm, 0 otherwise. <i>Data Source:</i> Best's Key Rating Guide                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Return on assets      | After-tax net income/assets. <i>Data Source:</i> NAIC database                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

**Table 2**  
Means, standard deviations and correlations (N = 3861)

| Variable                 | Mean   | S.D.   | 1       | 2       | 3       | 4       | 5       | 6       | 7       | 8       | 9      | 10      | 11      | 12      | 13      | 14     | 15      | 16 |
|--------------------------|--------|--------|---------|---------|---------|---------|---------|---------|---------|---------|--------|---------|---------|---------|---------|--------|---------|----|
| 1. Product variety       | 0.35   | 0.28   | 1       |         |         |         |         |         |         |         |        |         |         |         |         |        |         |    |
| 2. Geographic dispersion | 0.41   | 0.37   | .289**  | 1       |         |         |         |         |         |         |        |         |         |         |         |        |         |    |
| 3. Group affiliation     | 0.66   | 0.47   | .158**  | .258**  | 1       |         |         |         |         |         |        |         |         |         |         |        |         |    |
| 4. Third party ratings   | 236.45 | 152.96 | -.142** | -.101** | -.148** | 1       |         |         |         |         |        |         |         |         |         |        |         |    |
| 5. Cultural distance     | 0.01   | 0.15   | .043**  | .040*   | .016    | -.028   | 1       |         |         |         |        |         |         |         |         |        |         |    |
| 6. Foreign dummy         | 0.01   | 0.11   | .081**  | .065**  | .064**  | -.049** | .640**  | 1       |         |         |        |         |         |         |         |        |         |    |
| 7. Age                   | 48.61  | 42.38  | .320**  | -.016   | -.130** | .054**  | .017    | -.014   | 1       |         |        |         |         |         |         |        |         |    |
| 8. Size                  | 2.33   | 0.78   | -.238** | -.308** | -.499** | .113**  | -.047** | -.090** | .047**  | 1       |        |         |         |         |         |        |         |    |
| 9. Business line         | 0.31   | 0.46   | .386**  | .145**  | .039*   | -.059** | .031    | .049**  | .173**  | -.100** | 1      |         |         |         |         |        |         |    |
| 10. Investment           | 0.12   | 0.16   | .089**  | .029    | -.056** | .120**  | -.034*  | -.032*  | .320**  | .099**  | .047** | 1       |         |         |         |        |         |    |
| 11. Price                | 0.02   | 0.03   | -.106** | .010    | -.068** | .178**  | -.006   | -.010   | -.048** | .083**  | .033*  | .033*   | 1       |         |         |        |         |    |
| 12. Leverage             | 1.67   | 1.14   | .099**  | .149**  | .094**  | -.468** | .011    | .019    | -.080** | -.139** | .128** | -.209** | -.108** | 1       |         |        |         |    |
| 13. Expense ratio        | 31.00  | 18.69  | .043**  | .011    | -.118** | .215**  | .013    | -.022   | .122**  | .154**  | -.003  | .122**  | .174**  | -.152** | 1       |        |         |    |
| 14. International ratio  | 0.00   | 0.04   | .030    | .062**  | .030    | .042**  | .109**  | .041*   | -.003   | -.051** | .001   | .018    | .016    | -.011   | .015    | 1      |         |    |
| 15. Organization         | 0.70   | 0.46   | -.088** | .268**  | .306**  | -.105** | .049**  | .076**  | -.560** | -.250** | -.021  | -.288** | .054**  | .118**  | -.057** | .057** | 1       |    |
| 16. Distribution system  | 0.63   | 0.48   | .239**  | -.011   | -.020   | -.042** | -.009   | -.015   | .293**  | -.007   | .209** | .089**  | .009    | .011    | .152**  | .008   | -.078** | 1  |

\*Correlation is significant at the 0.05 level (two-tailed), \*\*correlation is significant at the 0.01 level (two-tailed).

**Table 3**

Summary of regression results with performance as dependent variable

| Independent variable    | Model 1   | Model 2   | Model 3   | Model 4   |
|-------------------------|-----------|-----------|-----------|-----------|
| Product variety         |           |           |           | -.053***  |
| Geographic dispersion   |           |           |           | -.031***  |
| Group affiliation       |           |           |           | .020***   |
| Third party ratings     |           |           |           | .228***   |
| Cultural distance       |           |           | .011      | .011      |
| Foreign dummy           |           | -.015**   | -.022**   | -.012     |
| Age                     | -.034***  | -.031***  | -.031***  | -.011     |
| Size                    | -.008     | -.008     | -.008     | -.019**   |
| Business line           | -.040***  | -.038***  | -.038***  | -.024***  |
| Investment              | .043***   | .042***   | .042***   | .046***   |
| Price                   | .039***   | .036***   | .036***   | .009      |
| Expense ratio           | -.846***  | -.859***  | -.859***  | -.756***  |
| Leverage                | .026***   | .024***   | .024***   | -.002     |
| International ratio     | .006      | .006      | .005      | -.002     |
| Organization            | .019**    | .019**    | .019**    | .039***   |
| Distribution system     | -.017**   | -.017**   | -.016**   | .001      |
| Adjusted R <sup>2</sup> | .755      | .777      | .777      | .821      |
| F-Value                 | 851.28*** | 897.93*** | 841.92*** | 884.22*** |

\*\*\* $p < .01$ , \*\* $p < .05$ , \* $p < .1$ . All the independent variables were mean centered in the regression models to remove non-essential multicollinearity. All models were run with 4-year dummies (2004 dummy is excluded). All beta values reflected standardized beta values.

three regression models (Models 1–3, Table 3). Model 1 is the base model or control, wherein we run the model with the control variables alone. Model 2 includes all the variables of Model 1 along with the foreign dummy. In Model 2 the foreign dummy term loaded negatively ( $\beta = -.015$ ,  $p < .05$ ) with respect to performance, indicating that foreign firms underperformed domestic firms. Model 3 attempts to verify if the performance differential exists even after controlling for cultural distance. In this model the negative loading of the foreign dummy term increased marginally ( $\beta = -.022$ ,  $p < .05$ ), validating the notion that performance differences exists across foreign and domestic firms even after controlling for cultural distance. Model 4 adds the four boundary spanning variables to Model 3. It should be noted that no specific hypotheses were offered on the influence of each of these boundary spanning variables on performance. In Model 4, each of these variables were significant with beta values of  $-.053$ ,  $-.031$ ,  $.020$ , and  $.228$  (respectively,  $p < .01$ ), validating the importance of these boundary spanning roles. Importantly, to our surprise we found that foreign dummy became insignificant ( $\beta = -.012$ , non-significant) in Model 4.

Four regression models were run to test the four hypotheses proposed in the study, and the results are reported in Table 4. Hypothesis 1 proposed that foreign firms will operate in more product lines than domestic firms. Findings indicate clear support for this hypothesis with a positive loading by the foreign dummy term ( $\beta = .069$ ,  $p < .01$ ). Hypothesis 2 proposed that foreign firms will have a greater spread in their operations compared to domestic firms. This hypothesis was not supported even though the beta loading was in the direction proposed ( $\beta = .021$ , non-significant). It should be noted there was no support either for the counter argument that foreign firms will concentrate on fewer locations. Hypothesis 3 proposed that foreign firms will be

**Table 4**  
Hypothesis testing—summary of regression results

| Independent variable    | Dependent variable |                       |                   |                     |
|-------------------------|--------------------|-----------------------|-------------------|---------------------|
|                         | Product variety    | Geographic dispersion | Group affiliation | Third party ratings |
| Foreign dummy           | .069***            | .021                  | .030**            | -.028**             |
| Cultural distance       | -.027              | -.008                 | -.031             | -.007               |
| Age                     | .250***            | .132***               | -.006             | -.011               |
| Size                    | -.206***           | -.233***              | -.442***          | .010                |
| Business line           | .288***            | .101***               | -.006             | .019                |
| Investment              | .020               | .114***               | .058***           | -.002               |
| Price                   | -.064***           | .023                  | -.038***          | .111***             |
| Expense ratio           | .056***            | .115***               | .010              | -.427***            |
| Leverage                | .049***            | .060***               | -.035***          | .135***             |
| International ratio     | .019               | .031**                | -.002             | .040***             |
| Organization            | .014               | .299***               | .207***           | -.062***            |
| Distribution system     | .097***            | -.070***              | -.004             | -.065***            |
| Adjusted R <sup>2</sup> | .281               | .186                  | .287              | .259                |
| F-Value                 | 95.066***          | 56.265***             | 98.326***         | 85.186***           |

\*\*\* $p < .01$ , \*\* $p < .05$ , \* $p < .1$ . All the independent variables were mean centered in the regression models to remove non-essential multicollinearity. All models were run with 4-year dummies (2004 dummy is excluded). All beta values reflected standardized beta values.

affiliated with groups in greater numbers than domestic firms. This hypothesis was statistically supported ( $\beta = .03$ ,  $p < .05$ ). Hypothesis 4 proposed that foreign firms will have higher levels of third party ratings compared to domestic firms. Contrary to Hypothesis 4, foreign firms have lower third party ratings, indicated by the negative loading of the foreign dummy term ( $\beta = -.028$ ,  $p < .05$ ).

To increase confidence in the study findings, we also conducted additional analysis, repeating the analysis reported in Tables 3–5 using a larger sample of foreign firms. While this study's focus is on firms who are 100% foreign owned, we relaxed the sample requirement to include any firms which claim any foreign affiliation. This increased the sample size of foreign firms to 399 firms from 13 countries. To reduce confounds, we had dropped these firms from the study sample, as many of these firms either did not disclose the extent of foreign ownership or had less than 100% foreign ownership. Findings indicate the earlier pattern of the results to be stable and consistent, even though there were variations in the beta loading and  $R^2$  values (typically higher in many instances). Since there were no inferential differences compared to what is already reported, they are not summarized in this paper.

#### 4.1. Post hoc analysis

We were intrigued by the fact that, when the four dependent variables were included in Model 4 (Table 3), foreign dummy becomes insignificant. When this result is taken together with the four regression models which test the hypotheses in Table 4, one could infer that boundary spanning variables potentially mediate the relationship between foreign ownership and performance (Baron & Kenny, 1986). This served as a motivator to conduct additional analysis for mediation. Conventional practice in such circumstances is to use simple mediation models, and Preacher and Hayes (2007) offer four reasons why multiple

mediation models should be used<sup>5</sup>. First, this approach is analogous to testing using regression with several predictors, to determine the overall effect. Second, in multiple mediation models, the mediation effect is captured after controlling for other mediators and is therefore more reasonable. Third, contingent on situation, simple mediation models can lead to biased results due to omitted variable problems. Fourth, having several variables allows one to determine the relative influence of one variable over another. However, they acknowledge that the preference for single mediator models in many instances is driven by the fact that it is simpler to test. Therefore, as a next step we tested for multiple mediation based on the methodology suggested. The results of multiple mediation models are presented in Table 5.

The multiple mediation model was supported with an adjusted R-square value of .8035 ( $p < .01$ ). Consistent with the findings of the ANOVA and regression models (reported earlier), findings of the multiple mediation models (see Table 5) indicate that foreign firms underperform domestic firms when the mediators are not factored in ( $\beta = -.2041$ ,  $p < .05$ ). When the multiple mediators are factored in, namely the direct effect, we find no performance differences ( $\beta = -.1109$ , non-significant) confirming the earlier suspicion of mediation. Table 5 also presents the results of the influence of the independent variable (i.e. foreign dummy) on the mediators (i.e., boundary spanning variables) (Path A), and the influence of mediators of performance (Path B) along with the partial effect of control variables. Findings indicate that foreign firms have greater product variety and group affiliation ( $\beta = .6084$  and  $.2624$ ,  $p < .01$  and  $.05$ , respectively), which is supportive of Hypotheses 1 and 3. Geographic dispersion received no statistical support, although the loadings were in the direction stated in Hypothesis 2, while foreign firms had lower third party ratings ( $\beta = -.2477$ ,  $p < .05$ ), contrary to Hypothesis 4. While the consistency in results reported in Tables 3 and 4 with Table 5 serve as a good check of the robustness of this paper's finding, readers are advised to use Table 5 results for inferential purposes, as it incorporates the varied relationships in the statistical models.

While no hypothesis was offered on the influence of these boundary spanning variables on performance, multiple mediation models do show the effect of the variables on performance (Path B). Product variety and geographic dispersion negatively influence performance ( $\beta = -.0553$  and  $-.0318$ ,  $p < .01$ ), while group affiliations and third party ratings positively influence performance ( $\beta = .0218$  and  $.2400$ ,  $p < .01$ ). In line with results in Table 3 (Model 4), only five of the eleven control variables were significant. In terms of partial effects, cultural distance, age, price, expense ratios, international ratio and distribution system did not impact performance while size, business line, invest-

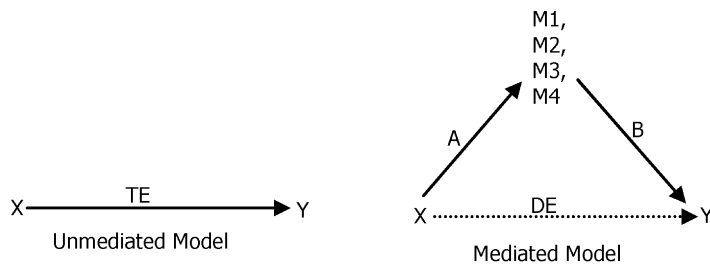
<sup>5</sup> In similar instances there is also a predisposition by many researchers to use path models. However, Preacher and Hayes (2007) severely caution against such a choice, as these models require setting the mediators' residual covariance to zero, something which cannot be defended either theoretically or based on statistical properties in instances such as this study.



**Table 5**  
Summary of multiple mediation model (dependent variable: performance)

| Independent variable    | Total effect on performance (TE) <sup>a</sup> | Direct effect on performance (DE) <sup>b</sup> | Foreign dummy to mediators (Path A) <sup>c</sup> | Direct effects of mediators on performance (Path B) <sup>d</sup> | Partial effect of control variables on performance |
|-------------------------|-----------------------------------------------|------------------------------------------------|--------------------------------------------------|------------------------------------------------------------------|----------------------------------------------------|
| Foreign dummy           | –.2041**                                      | –.1109                                         |                                                  |                                                                  |                                                    |
| Product variety         |                                               |                                                | .6084***                                         | –.0553***                                                        |                                                    |
| Geographic dispersion   |                                               |                                                | .1852                                            | –.0318***                                                        |                                                    |
| Group affiliation       |                                               |                                                | .2624**                                          | .0218***                                                         |                                                    |
| Third party ratings     |                                               |                                                | –.2477**                                         | .2400***                                                         |                                                    |
| Cultural distance       |                                               |                                                |                                                  |                                                                  | .0120                                              |
| Age                     |                                               |                                                |                                                  |                                                                  | –.0133                                             |
| Size                    |                                               |                                                |                                                  |                                                                  | –.0211***                                          |
| Business line           |                                               |                                                |                                                  |                                                                  | –.0247***                                          |
| Investment              |                                               |                                                |                                                  |                                                                  | .0470***                                           |
| Price                   |                                               |                                                |                                                  |                                                                  | .0101                                              |
| Leverage                |                                               |                                                |                                                  |                                                                  | –.7367***                                          |
| Expense ratio           |                                               |                                                |                                                  |                                                                  | –.0010                                             |
| International ratio     |                                               |                                                |                                                  |                                                                  | –.0020                                             |
| Organization            |                                               |                                                |                                                  |                                                                  | .0403***                                           |
| Distribution system     |                                               |                                                |                                                  |                                                                  | .0017                                              |
| Adjusted R <sup>2</sup> |                                               |                                                | .8035***                                         |                                                                  |                                                    |
| F-Value                 |                                               |                                                | 790.012***                                       |                                                                  |                                                    |

\*\*\* $p < .01$ , \*\* $p < .05$ , \* $p < .1$ . All the independent variables were mean centered in the regression models to remove non-essential multi-collinearity. All models were run with 4-year dummies (2004 dummy is excluded). All beta values reflected standardized beta values.



<sup>a</sup> Total effect (TE) is the influence of independent variable (IV) on dependent variable (DV) in the unmediated model.

<sup>b</sup> Direct effect (DE) is the influence of IV on DV in the mediated model.

<sup>c</sup> Influence of IV on mediator in the mediated model (Path A).

<sup>d</sup> Influence of Mediator on DV in the mediated model (Path B).

ments, leverage and organization impact performance after controlling for the influence of the mediators. While the negative impact of size (beta =  $-.0211$ ,  $p < .01$ ) on performance may be surprising, this finding is not entirely inconsistent with extant LOF research. For instance, Mezas (2002a, 2002b) reported a negative relationship between size and LOF. One possible untested explanation for these results could be due to proliferation of procedures/process and structural inertia patterns commonly attributed to such firms (Hannan & Freeman, 1984).

## 5. Discussion

The existence of LOF implies that foreign firms suffer disadvantages in alien environments, yet it is not surprising to see foreign firms thriving in many alien markets. This paper's goal was to understand how foreign firms minimize the effects of LOF and survive in an alien market despite LOF. To this extent, we do find that, on average, foreign firms underperform domestic firms, which is in line with previous research findings. Once boundary spanning variables are incorporated, performance differences between foreign and

domestic firms become insignificant in step-wise regression and multiple mediation models. Findings support the overall theoretical premise of this paper that foreign firms will undertake strategic postures to enhance their ability to gain information from the environment and transmit favorable information. Under this framework, four hypotheses were proposed, and findings support two of the four strategies suggested. For the remaining two hypotheses, no statistical support was found for one, and in the case of the other, findings were quite opposite to what was proposed.

Using extant research, Hypothesis 2 suggested that foreign firms would operate in greater geographic areas in host country markets than domestic firms to allow tapping of local information, minimize region specific risk, gain economies of scale, and increase legitimacy. No statistical support was found for this assertion or for the contrary position that increased geographic exposure for foreign firms may increase LOF. Mezas (2002, p 240) provides one reason as to why foreign firms may be reluctant to spread operations. In his study, he asserts that direct experience in one state may enhance operational effectiveness in that particular state, but not necessarily in other states due to

variations in state laws. Similar situations can arise in the insurance sector, as insurance firms are regulated at the state level with prior and concurrent approvals required in many operational facets (e.g., pricing). This mandate requires insurance firms to adapt according to each state's regulations and laws, adding complexity to foreign firms. Since these isomorphic pressures vary by state, foreign firms need to quickly understand and adopt appropriate practices in these micro-institutional environments, greatly magnifying the issue of liabilities of foreignness for these firms. Therefore by focusing on fewer markets, these firms can reduce within-host country variability as well as regulatory filings, while potentially operating with simpler organizational structures. Surveys on foreign investments in the United States also find distinct regional patterns in concentration (Shannon & Zeile, 1999). Hence in retrospect, one may infer that foreign firms are better off by limiting exposure to certain geographic segments within the United States despite the trade-off of this choice for boundary spanning roles.

**Hypothesis 4** proposed that foreign firms would use institutional legitimating devices to transmit favorable information to enhance reputation and trustworthiness in an alien environment. To the contrary, the findings were quite opposite to the direction proposed, wherein foreign firms compared poorly to domestic firms. This finding leads us to speculate as to why this would be the case. One simplistic or naive explanation is that these foreign firms, being poor financial performers relative to their domestic counterparts, are unable to improve their CARs. This explanation does not bode well for foreign firms, as they could face a downward spiral in ratings, possibly leading to bankruptcy, making this explanation less plausible. Moreover, group affiliation has a positive relationship with performance, further invalidating this argument. Another potential explanation could be that these firms are conforming to their home countries' "success recipes" and thereby follow differing strategies compared to their domestic counterparts (Elango & Sethi, 2007). We found some anecdotal support of this notion in our focus group discussions with industry executives of foreign firms. These executives mentioned that such ratings (of any kind) are not yet the norm in their home countries (i.e., Switzerland, Germany, and Denmark) and acknowledged that they find it hard to factor in such ratings as much as they should. This should not be a total surprise, as previous studies in other contexts have indicated that financial ratios of multinational firms do not conform to industry norms even in global industries (Duysters & Hagedoorn, 2001). More importantly, Miller and Eden (2006) show that such strategic conformance with the local environment led to higher performance only in circumstances wherein there were fewer competitors. By making such a choice, however, foreign firms are forced to resist isomorphic pressures of the local environment (Kostova & Zaheer, 1999).

## 6. Conclusion

This study began with the objective of understanding the response strategies of foreign firms facing LOF in an

alien environment. Study findings support the overall rationale of this paper that foreign firms adapt through differential strategic posture to overcome disadvantages of LOF. To this extent, this study extends the scope of the LOF literature using a sample of service sector firms operating in a regulated industry. Specifically, we find that in order to cope with LOF, foreign firms tend to have greater product variety and are likely to be affiliated with a business group, compared to domestic firms. Contrary to our expectations, foreign firms have lower third party ratings when compared with domestic firms. We conclude this paper by commenting on the managerial implications of this study as well as its limitations.

### 6.1. Managerial implications

The findings of this study offer implications for managers of foreign firms seeking to reduce LOF in alien markets. First, evidence supports the notion that foreign firms offer greater product variety, thus increasing the chances of offering a desirable product in the alien environment. Supplemental findings indicate that product variety by itself is negatively related to performance. It is quite plausible that firms (domestic and foreign) might stretch themselves too much by having too many products, leading to poor performance (Hitt et al., 1994). Therefore, while foreign firms need greater product variety to increase the chances of success and increase awareness and acceptability in host nations, it might be desirable to prune product lines which are not profitable after the foreign firm has established itself. Additionally, reducing product variety will help reduce firm size, which also seems desirable, as findings indicate that size is negatively related to performance.

Second, for foreign firms entering host countries, the results of **Hypotheses 1 and 2** taken together present an interesting strategy to follow. This strategy offers a good compromise between the tradeoffs in reducing LOF and operating efficiently. In this strategy, foreign firms offer greater product variety but concentrate in selected geographic markets with maximum potential (i.e., population centers) in host nations. By doing so, these firms limit the risks of LOF since they have an increased focus on and a better understanding of customers in specific regions in an alien market, while at the same time gaining from operational scale and scope benefits through sales generated from a large variety of products.

Third, findings also supported the performance benefits of such group affiliations validating the efficacy of networks (i.e., groups) for information assimilation and dissemination (Hargadon & Sutton, 1997). Therefore foreign firms are better off being affiliated with groups. These groups reduce unfamiliarity hazards by quickly internalizing information about the external environment and allowing for gains from "legitimacy spillovers". Managers may want to ensure that such membership optimizes knowledge acquisition, connections, resources and operational efficiency. Fourth, the importance of institutional legitimating devices as a means to enhance reputation and trustworthiness in an alien environment was highlighted. Apparently foreign firms have problems adapting to local norms which are based on

financial ratios due to reasons discussed earlier. In this case, it leads to the related question what should foreign firms do in such instances to gain legitimacy. One suggestion for foreign firms is that they could seek institutional legitimating devices such as customer satisfaction ratings or complaints, wherein host country operational norms do not bias the results against foreign firms.

## 6.2. Limitations

This study relied on large samples based on publicly available secondary data and therefore has its design limitations. First, it focused on one type of industry to reduce the chance that industry confounds would invalidate the findings. Therefore, its findings should be interpreted within its context. Second, due to this study's reliance on secondary data, its empirical boundaries are limited by the availability of data. For instance, entry mode choices made by a firm or firm-specific competencies which attenuate the effects of LOF are not factored in the analysis. Third, group affiliation is measured using a dummy variable due to constraints of data availability. The usage of dummy variable results in this study does not allow for differentiating across variations in groups. Finally, while this study presented supplemental analysis on performance, such outcomes are dependent on a large number of temporal and dynamic issues which cannot be covered in a cross-sectional study such as this. Therefore, conventional caveats apply to this study. However, duplication of this study's results with other industry samples, differing time periods, and diverse country contexts will alleviate some of the limitations.

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